

CASE STUDY

CARTER

RACING

SURVIVORSHIP BIAS & DECISION-MAKING UNDER UNCERTAINTY



BUSINESS CONTEXT



**FIRST SEASON
HIGH FINANCIAL PRESSURE
NATIONAL TV EXPOSURE
ONE CRITICAL DECISION**



DATA OVERVIEW



OBSERVED DATA

24 RACES

7 ENGINE FAILURES

29% FAILURE RATE

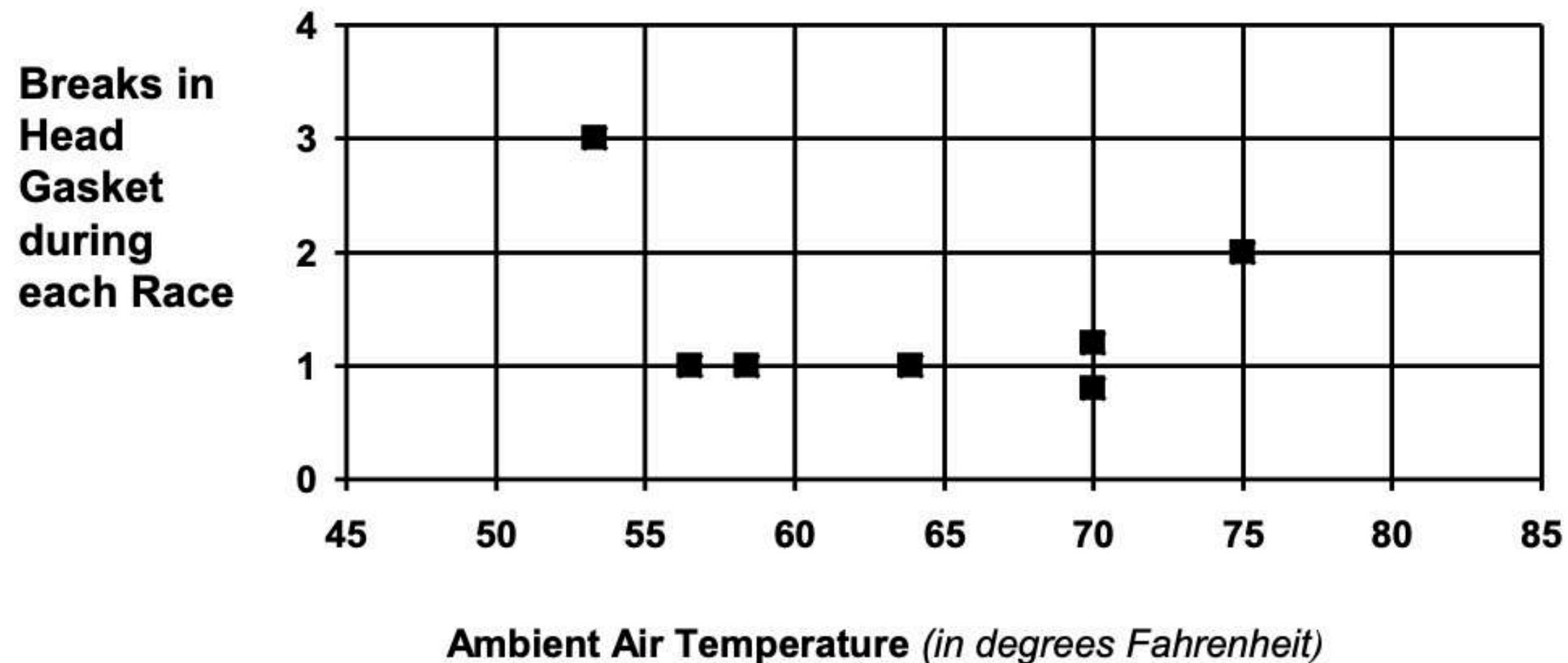
PERFORMANCE VS FAILURE



62.5% RACES FINISHED
80% TOP-5 WHEN FINISHED
4 WINS

TEMPERATURE & DATA LIMITS

Relationship between Temperature and Gasket Failures



**OBSERVED RANGE: 53–
82°F**
**NO CORRELATION IN
RANGE**
TODAY: 40°F

SURVIVORSHIP BIAS



SURVIVORSHIP BIAS
FOCUS ON WINS
FAILURES UNDERWEIGHTED
OVERCONFIDENCE

WHY FAILURE DOMINATES THE DECISION

IF FAILURE OCCURS:

- ENGINE LOSS
- SPONSOR LOSS
- REPUTATION DAMAGE
- SURVIVAL RISK



STRATEGIC VIEW (BCG MATRIX)

POCONO RACE = QUESTION MARK
HIGH OPPORTUNITY
HIGH UNCERTAINTY
WEAK RELIABILITY



BCG Growth Share Matrix



WE RACE OR WE REGRET

- COLD CAN BE MANAGED — WE'VE WON IN THE 50s.
- NO RACE = NO SPONSORS, NO GROWTH, ALL DEBT.
- FINISH = GOODSTONE \$2M, OIL DEAL, NEXT SEASON SECURED.
- WE RACE TO WIN, NOT TO WONDER.

START THE ENGINE



ANALYTICAL DECISION

BEST DECISION:

- AVOID IRREVERSIBLE LOSS
- PRESERVE FUTURE OPPORTUNITIES
- IMPROVE PREDICTIONS LATER

→ RACE OR NOT RACE?

SLIDE 10 — FINAL CONCLUSION

**GOOD ANALYTICS PROTECTS SURVIVAL BEFORE CHASING
PERFORMANCE**

THANK YOU

